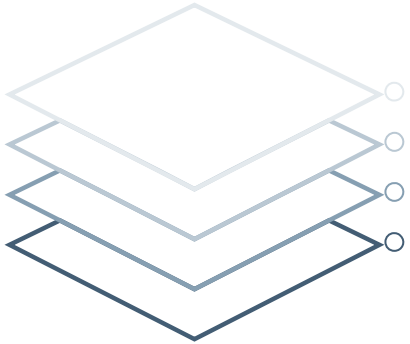
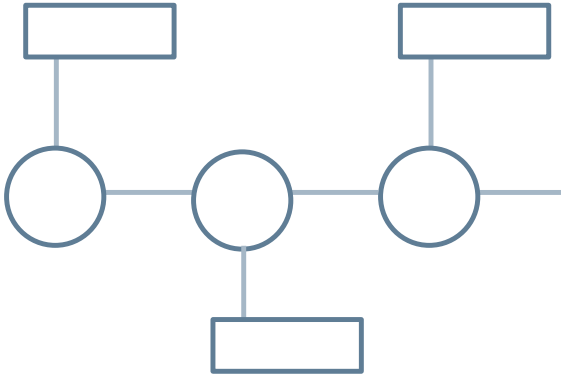




ClearSight AI

Illuminating the World, One Pixel at a Time



Presented by Vihaan Mishra

What is AI?

- Computer systems/networks that think like humans

- Makes data-driven decisions



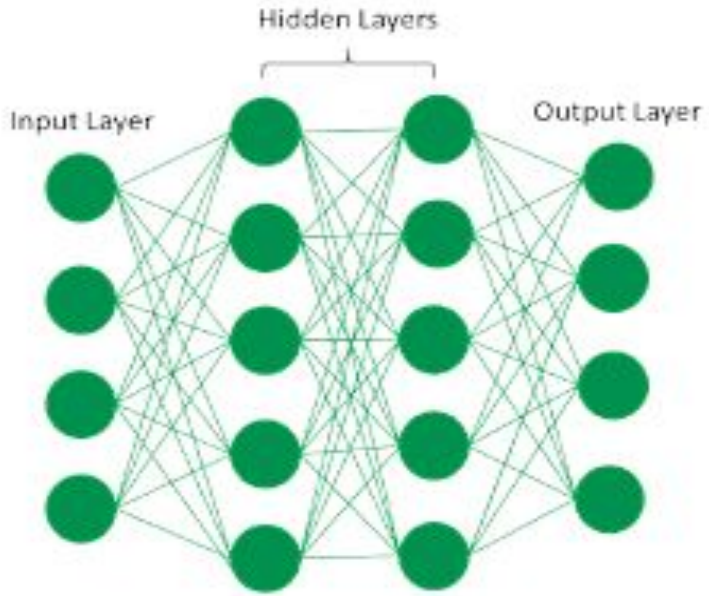
Main Concept in AI - **Neural Networks**

Works like our own brain

Input layer - takes in raw data

Hidden layers - series of nodes that work to detect patterns

Output layer - makes the prediction



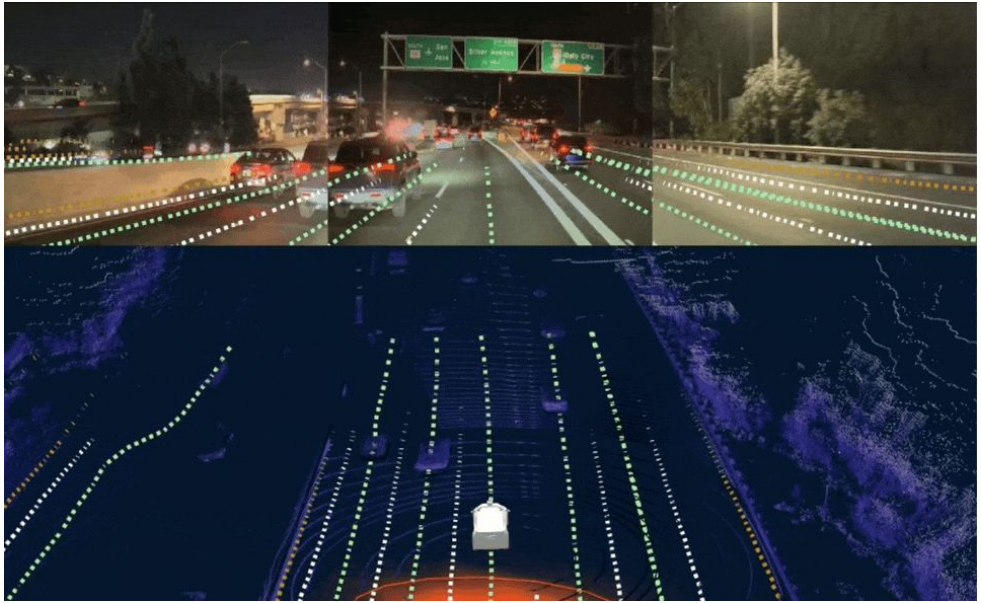
Inspiration

Self-Driving vehicles

- Safety for pedestrians and cyclists

Environmental Monitoring

- Process low-light wildlife camera traps or nighttime satellite/aerial imagery to monitor ecosystems
- Track endangered species

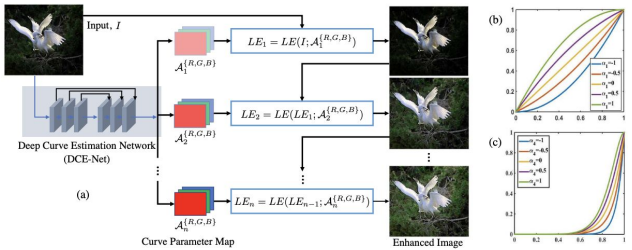




Research Question

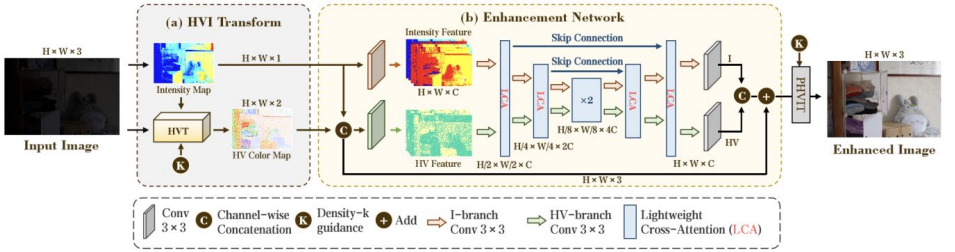
Can combining different parts of existing models result in improved outcomes in the field of low-light image enhancement?

Zero-DCE



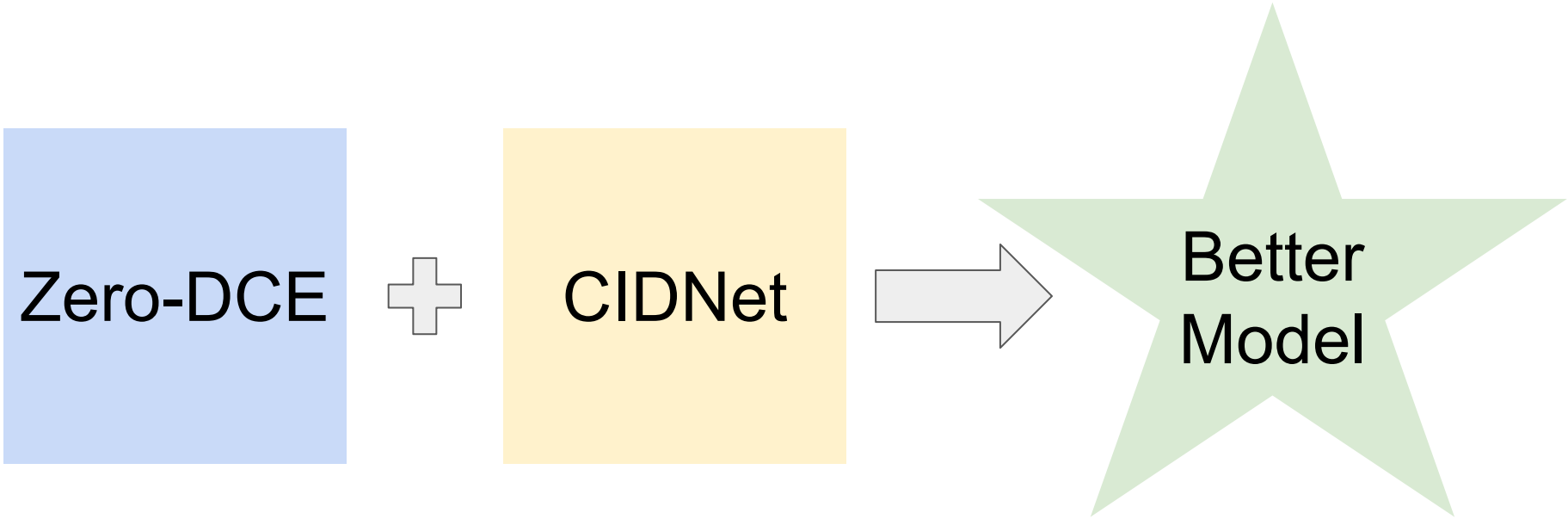
+

CIDNet



= ?

In short:



Research Process

Key points:

- ReLU Activation and Tanh - Computationally efficient & maintains complexity of images
- Extra layers created from early convolutions are brought into later convolutions to maintain features
- Run for 250 epochs

Frameworks used:

PyTorch

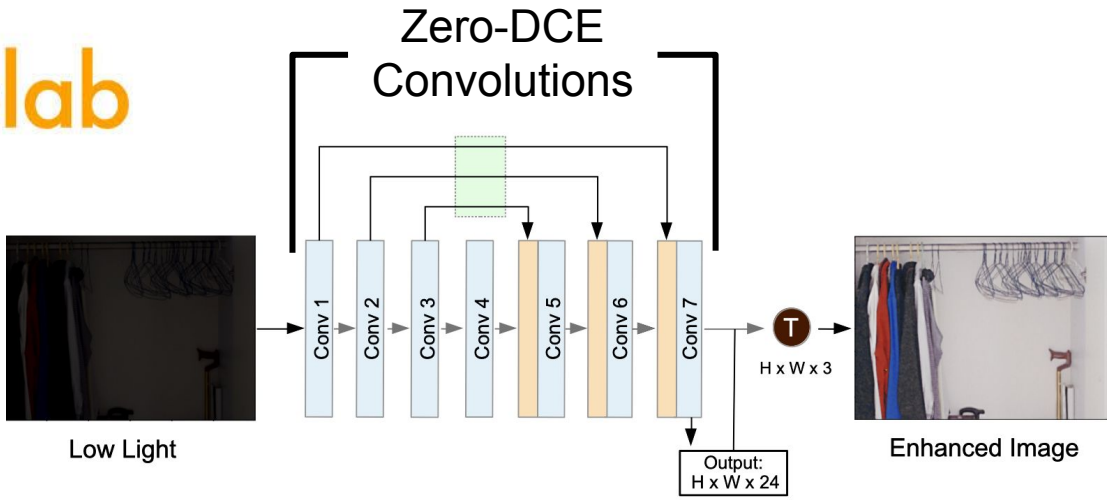
pillow

python + colab

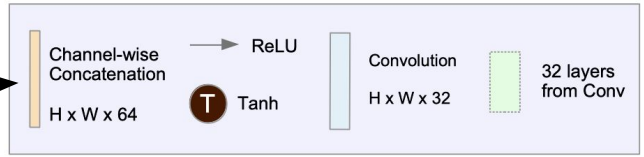
pandas

NumPy

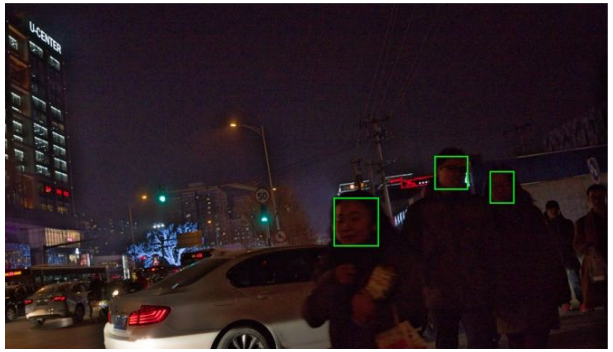
matplotlib

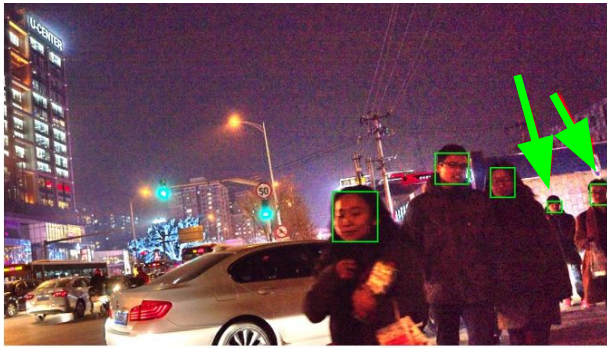


CIDNet Concatenation



Data and Results





Input

Output

Fig 1. Comparison between low-light image and processed image after going through Clear Sight AI

Model	PSNR (dB) ↑	SSIM ↑
ClearSight AI	**22.51**	**0.791**
CIDNet	22.40	0.785
Zero-DCE++	19.80	0.720

Fig 2. Table showing metrics comparing ClearSight AI versus CIDNet and Zero-DCE

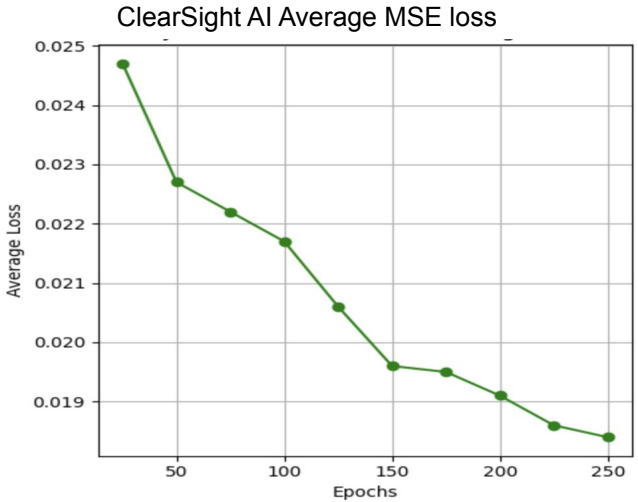


Fig 3. Graph of average loss every 25 epochs

Visual Output from ClearSight AI

Input Image

ClearSight AI

LOL Dataset



Input Image

ClearSight AI

Dark Face Dataset



Why ClearSight AI is a Winner

1. Context-Aware Enhancement

2. Proven Superior on Dark Face Dataset

3. Applications in the Real World

Thank You for Listening!

Special thanks to
Dr. Bhargava and Dr. Lulla
for organizing this
presentation!

Do you have any Questions?